

Sohum Sharma

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Education

PES University, Computer Science

2022-2026

CBSE Class XII Modern School, Barakhamba Road, New Delhi

2019

Experience

Nimble Vision, Intern [Personal Project]

February 2025

- Trained a custom YOLOv8 model to detect pets, humans, and trucks for use in surveillance camera systems, achieving 72% accuracy.
- Leveraged Qualcomm AI Hub to compile, optimize, and deploy the model efficiently on Samsung S22 series devices.
- Designed the solution with real-time performance and mobile deployment in mind for smart surveillance applications.

Nimble Vision, Intern [Team Project]

July 2024-August 2024

- Backend integration for IOT devices, achieved seamless improved system stability with LORAWAN network servers.
- Architected and implemented a scalable solution using Django, HTML, CSS, JavaScript.

Oil and Natural Gas Corporation(ONGC), Intern [Team Project]

May 2024-July 2024

- Automated payroll process for ONGC, significantly reduced manual effort, minimized errors, and enhanced timely, accurate payroll delivery.
- Designed and implemented a streamlined system using SAP ABAP to automate data retrieval, processing, and pay slip pdf generation.

Projects

Detection of Multiple Neurological Disorders using Machine Learning and Deep Learning(Ongoing Project)

- Developing a machine learning and deep learning multi-modal model to detect neurological disorders (Alzheimer's, Dementia, Parkinson's) using MRI, PET, and OCT images, annotated using Roboflow.
- Achieved 98 % accuracy with a ResNet model trained on 10,000 MRI images; compared CNN and Transformer architectures with different fusion methods (early, intermediate, late) to identify optimal performance.
- Integrated YOLOv8 for precise region detection and going to utilized LIME and SHAP for clinical interpretability.

BERT+ResNet based Multimodal Emotion Detection

- Developed a multimodal deep learning model combining textual embeddings (BERT) and visual features (ResNet50) to classify emotions.
- Achieved 68% accuracy by extracting text embeddings from sentences and visual features directly from video frames.
- Enhanced prediction performance by effectively integrating textual and visual data into a unified end-to-end model.

BERT-based Question Answering System

- Built a question-answering system using a BERT-based model trained on the CoQA dataset for accurately extracting answers from text.
- Transformed and tokenized data into a SQuAD-like format, implementing custom PyTorch data loaders and fine-tuning BERT using Hugging Face's Transformers library.
- Integrated experiment tracking with Weights & Biases (wandb) to monitor training progress, achieving improved model accuracy through effective hyperparameter tuning.

Autonomous Waste Monitoring with Active Learning

- Built an IoT system using Python, scikit-learn, and IR sensors to predict industrial bin fill status and created artificial data for training the model.
- Created a fully-automated active learning loop that uses prediction confidence (95%) to self-label new data, enabling the model to continuously improve without human supervision.
- Engineered a dual-sensor array for reliable data capture and trained a RandomForestClassifier for the prediction task.

Full-Stack Concert Ticket Booking System

- Developed a full-stack web application for booking concert tickets, providing a smooth and efficient user experience.
- Built the backend using Flask (Python) and handled database management with MySQL.
- Designed a clean and user-friendly frontend using HTML and CSS for easy customer interaction.

Courses

- [Generative AI for Everyone\(Open AI\)](#)
- [Generative AI with LLMs\(Open AI\)](#)
- [How Diffusion Models Work\(Open AI\)](#)
- [Neural Networks and Deep Learning\(Open AI\)](#)
- [Mastering Excel for Data Analytics from NSIC with A1 grade certificate](#)

Achievements

- [Google Agentic AI hackathon\(Got selected in the top 700 from 2000+ teams\)](#)